A smooth wiggle

Little drew a smooth line that rose and fell again. Big called it a sine wave. Could this shape help explain a speaker?

Slow at the turn

The swing slowed at each end and sped through the middle. A sine-like line could record its back-and-forth motion as time passed.

A bouncing spring

Big held a spring above a tray. Its end bounced up and down. Each bounce grew smaller until the spring settled.

Along the rope

Little moved a rope from side to side. A wiggle traveled toward Big, while each bit of rope moved back and forth.

Turning into power

A generator turns motion into electrical energy. As a magnet spins beside wire, the electric push grows, fades, and changes direction smoothly.

A speaker's rhythm

The speaker's cone moved in and out to make a steady tone. Little drew its smooth rhythm. Where else might a wiggle appear?

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

Why Electronics Use Sine Waves